

Generative AI and LLM Foundations Exercises

Learner workbook for PU B04 C01

This workbook tests whether the learner can connect technical mechanisms to evidence, evaluation and accountable institutional use. Complete the companion notebook before attempting Sections B and C.

Learning evidence

Area	Expected evidence
Conceptual	Accurate definitions and boundaries
Computational	Reproducible notebook results
Analytical	Reasoned comparison and interpretation
Governance	Controls, records, responsibility and escalation

Submission requirements

1. Answer every exercise in your own words.
2. Include notebook output or calculations where requested.
3. State assumptions and limitations.
4. Cite the evidence used for material claims.
5. Do not include credentials, personal data or confidential records.

Section A Foundations

Exercise 1 Concept distinctions

Explain the difference among a discriminative model, a generative model and a retrieval system. Give one institutional use for each.

Learner response:

Exercise 2 Autoregressive generation

Starting from the sequence-probability factorization, explain what a language model estimates at each generation step and why high probability does not establish truth.

Learner response:

Exercise 3 Tokenization

Tokenize one English and one French sentence using the notebook. Compare token counts, unknown-token behaviour and implications for multilingual service delivery.

Learner response:

Exercise 4 Context boundary

Design a context budget for summarizing a quarterly report containing a main report, annexes, metadata and an audit-status note. State what must never be silently truncated.

Learner response:

Section B Generation and Attention

Exercise 5 Decoding

Compare greedy, temperature, top-k and top-p decoding. Recommend and justify settings for a controlled first draft of an official narrative.

Learner response:

Exercise 6 Attention

Interpret the notebook attention matrix. State one valid conclusion and two conclusions the matrix cannot support.

Learner response:

Exercise 7 Groundedness

Decompose a generated financial summary into claims and design a claim-to-source verification table.

Learner response:

Section C Evaluation and Governance

Exercise 8 Evaluation

Develop an evaluation plan that separates fluency, groundedness, correctness, robustness, safety and decision fitness.

Learner response:

Exercise 9 Risk controls

Identify privacy, prompt-injection and excessive-authority risks in a retrieval-and-tool application. Assign one preventive and one detective control to each.

Learner response:

Exercise 10 Evidence chain

Specify the complete SRAI evidence record required before a generated output influences a consequential decision.

Learner response:

Section D Applied and Capstone Reasoning

Exercise 11 Official statistics

Explain why a numerically correct agricultural statistic may still be misleading. Identify all metadata that must travel with the value.

Learner response:

Exercise 12 Capstone pathway

Describe how Lessons 1-30 could progressively support construction of a small educational GPT-style system. Distinguish it from fine-tuning an open model and from building a frontier model.

Learner response:

Self assessment

Criterion	Not yet	Developing	Demonstrated
I distinguish probability from factuality	☐	☐	☐
I trace claims to evidence	☐	☐	☐
I explain decoding trade-offs	☐	☐	☐
I define accountable controls	☐	☐	☐